we know today are in reality models of some unknown reality. This is continuously noted when all of a sudden a new discovery seem to falsify an older model e.g. the current discussion of the possibility that neutrinos actually travel faster than the speed of light. If these experiments are true we have to remodel much of that which we today consider being the truth. Indirect model building is familiar to every chemist: although we cannot see the atoms and molecules which we add into a reaction vessel, we certainly have some idea about what is going to happen. It is indeed our enjoyment in developing models which causes us to experiment in science. We also want to be able to make quantitative predictions based on our models which we therefore formulate in mathematical terms. To allow tractable calculations most models involves simplifications of the "real world". Of course, since man is fallible some models may turn out to be wrong but as new data accumulate, wrong or naive models are replaced by better ones.

We have already shown how one model for the nuclear structure, the liquid drop model, has helped us to explain a number of nuclear properties, the most important being the shape of the stability valley. But the liquid drop model fails to explain other important properties. In this chapter we shall try to arrive at a nuclear model which takes into account the quantum mechanical properties of the nucleus based mainly on the interactions and properties of neutrons and protons. Naturally it is also possible to go even deeper and discuss the nuclear structure through the interactions of quarks or using the string theory but that is beyond the scope of this book and we instead refer to textbooks in nuclear physics.

6.1. REQUIREMENTS OF A NUCLEAR MODEL

Investigation of light emitted by excited atoms (J. Rydberg 1895) led N. Bohr to suggest the quantized model for the atom, which became the foundation for explaining the chemical properties of the elements and justifying their ordering in the periodic system. From studies of molecular spectra and from theoretical quantum and wave mechanical calculations, we are able to interpret many of the most intricate details of chemical bonding.

In a similar manner, patterns of nuclear stability, results of nuclear reactions and spectroscopy of radiation emitted by nuclei have yielded information which helps us develop a picture of nuclear structure. But the situation is more complicated for the nucleus than for the atom. In the nucleus there are two kinds of particles, protons and neutrons, packed close together, and there are two kinds of forces - the electrostatic force and the short range strong nuclear force. This more complex situation has caused slow progress in developing a satisfactory model, and no single nuclear model has been able to explain all the nuclear phenomena.

6.1.1. Some general nuclear properties

Let us begin with a summary of what we know about the nucleus, and see where that leads us.

In Chapter 4 we observed that the average binding energy per nucleon is almost constant for the stable nuclei (Fig. 4.3) and that the radius is proportional to the cube root of the mass number. We have interpreted this as reflecting fairly uniform distribution of charge and mass throughout the volume of the nucleus. Other experimental evidence supports this interpretation (Fig. 4.4). This information was used to develop the liquid drop model, which successfully explains the valley of stability (Fig. 4.1). This overall view also supports the assumption of a strong, short range nuclear force.

A more detailed consideration of Figures 4.1 and 4.3 indicates that certain mass numbers seem to be more stable, i.e. nuclei with Z- or N-values of 2, 8, 20, 28, 50, and 82 (see also Table 4.1). There is other evidence for the uniqueness of those numbers. For example, if either the probability of capturing a neutron (the neutron capture cross-section) or the energy required to release a neutron is plotted for different elements, it is found that maxima occur at these same neutron numbers, just as maxima occur for the electron ionization energy of the elements He, Ne, Ar, Kr, etc. (i.e. at electron numbers of 2, 8, 16, 32, etc.). The nuclear N- or Z-values of 2, 8, 20, 28, 50, and 82 are called "magic numbers".

It seems logical that these magic numbers indicate some kind of regular substructure in the nucleus similar to the orbitals for the atomic electrons. Moreover, since the same magic numbers are found for the neutrons and for the protons, we would further assume

Table 6.1 Summary of the properties of the atomic constituents (independent movements in a central potential field).

Line property	Electron	Proton	Neutron
1 Mass (u)	0.000 5486	1.007 276	1.008 665
2 Charge (**e** units)	-1	$+1$	0
3 Spin quantum number s	1/2	1/2	1/2
4 Spin dipole moment	$\mu_e = 1$ B.m.	μ_p=2.793 n.m.	μ_n= 1.913 n.m.
5 Orbital quantum number	$0 \leq l \leq n-1$		$0 \leq l$
6 Permitted orbital field projections	$-\Sigma l \ldots + \Sigma l$		$-\Sigma l \ldots + \Sigma l$
7 Total angular quantum number	$(j = l \pm s)$		$j = l \pm s$
8 Total angular momentum	$(p_j = \Sigma\sqrt{(j(j+1))})$		$p_j = \Sigma\sqrt{(j(j+1))}$
9 Particle symbolism	nl^i		nl_j

Notes

2. The electron unit charge: $\mathbf{e} = 1.602 \times 10^{-19}$ C.

3. The spin angular momentum has the magnitude $p = \Sigma\sqrt{(s(s+1))}$, with the permitted projections on an external field axis: $\pm s\Sigma$ (for e, p, and n $= \frac{1}{2}\Sigma$).

7. l and s couple only in the one-electron system; see §11.2.4.

8. Permitted projections on an external field axis: Σm_j, where $j \leq m_j \leq j$.

9. n is the principal quantum number, l the azimuthal (orbital, radial) quantum number, i is the number of electrons in the particular n,l state, and j is the total angular momentum quantum number. For $l = 0$, 1, 2, etc., the symbols s, p, d, f, g, etc., are used.

that the neutrons and the protons build their substructure independently of each other, but in the same way. Another fact that must be indicative of the nuclear substructure is the stability for nuclei with even proton or even neutron numbers. Since we know that the individual nucleons have spin, we could postulate that nucleons in the nucleus must pair off with opposed spins.

6.1.2. Quantized energy levels

The nucleus would thus seem to consist of independent substructures of neutrons and protons, with each type of nucleon paired off as far as possible. Further, the nucleons are obviously grouped together in the magic numbers. From the decay of radioactive nuclei we know that the total decay energy (Q-value) of any particular nuclide has a definite value. Moreover, γ-emission from any particular nucleus involves discrete, definite values. These facts resemble the quantized emission of electromagnetic radiation (X-ray, UV, visible light, etc.) from atoms. We may conclude a similar explanation for the nucleus: decay of radioactive nuclei, whether α, β, or γ, involves a transition between discrete quantized energy levels.

6.1.3. The nuclear potential well

In our development of a model of the nucleus we have so far not considered the nuclear binding energy. Let us imagine the situation wherein a neutron of low kinetic energy approaches a nucleus (Fig. 6.1). Since the neutron is uncharged it is not affected by the Coulomb field of the nucleus and approaches the nucleus with no interaction, until it is close enough to experience the strong nuclear force F_n, which is always attractive, i.e. F_n is a positive quantity. At the point r_n the neutron experiences the strong attraction to the nucleus and is absorbed. The surface of the nucleus is assumed to extend from r_n to r_s since this distance represents the radius of the nucleus over which the nuclear force is constant. When the neutron is absorbed the nucleus is not organized optimally and the excess energy is released and emitted in the form of a γ-quantum. The energy of the γ-ray can be calculated from the known masses of the reactants and product nuclides: $E_\gamma = -931.5(M_{A+1} - M_A - M_n)$ The energy released is the neutron *binding energy* part of the total binding energy in the nucleus E_B. The total energy of the nucleus has thus decreased as is indicated in Figure 6.1(B); it is common to refer to this decrease as a *potential well*. The nucleons can be considered to occupy different levels in such a potential well. The exact shape of the well is uncertain (parabolic, square, etc.) and depends on the mathematical form assumed for the interaction between the incoming particle and the nucleus.

Protons experience the same strong, short range nuclear force interaction as they contact the nucleus. However, they also experience a long range repulsive interaction due to the Coulomb force between the positive incoming protons and the positive charge of the nucleus. This repulsion prevents the potential well from being as deep for

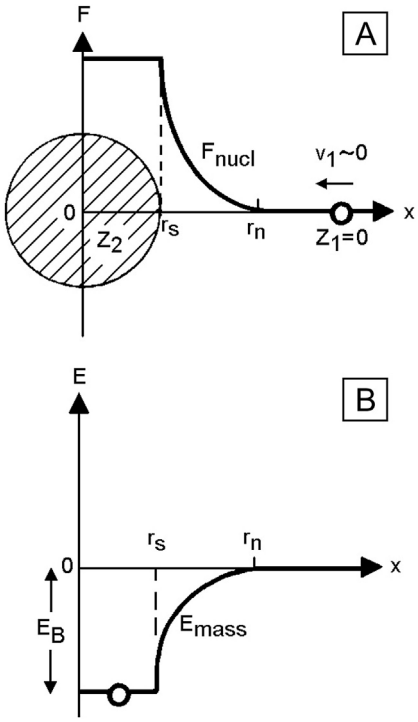


Figure 6.1 The approach along the x-axis of a neutron (charge $Z_1 = 0$, velocity $v_1 \approx 0$) towards a nucleus (charge Z_2; Fig. A) leading to the neutron being trapped at a certain energy level, E_B (Fig. B). F_{nucl} is the nuclear force extending from the surface of the "solid core" at r_s to the distance r_0. E_B is the nuclear binding energy in the potential well.

protons as for neutrons. Figure 6.2 shows the energy levels of the protons and of the neutrons in the nucleus of $^{116}_{50}$Sn.

6.2. ROTATIONAL ENERGY AND ANGULAR MOMENTUM

It is an intriguing fact that nucleons in nuclei, electrons in atoms, as well as large cosmic objects such as solar systems and even galaxies, are more dominated by rotational than by linear motion, although in our daily life the latter seems to be a more common phenomenon. In rotation there is a balance between two forces: the centrifugal force of inertia, which tries to move a body away from a center point, and an attractive force (gravitational, electrostatic, etc.), which opposes the separation.

In the preceding section we assumed that the nucleus existed in some kind of a potential well. One may further assume that a nucleon moves around in this well in a way not too different from the way the electron moves around the atomic nucleus, i.e. with an oscillation between kinetic and potential energy. With some hypothesis about the shape